

Git Hands-on Lab - 1

Aim:

To understand the basic Git workflow by configuring Git, setting Notepad++ as the default editor, creating a local repository, and pushing files to a GitHub repository.

Objectives:

- Configure Git with user details.
- Set Notepad++ as the default Git editor.
- Create a local Git repository.
- Add and commit files.
- Connect the local repository with GitHub.
- Push and pull files using Git.

Prerequisites:

- Windows Operating System
- Git Bash installed
- GitHub account
- Notepad++ installed
- Internet connection

Software Required:

- Git Bash
- GitHub
- Notepad++
- Windows Command Prompt

Procedure:

Step 1: Configure Git

Open Git Bash and execute the following commands.

```
git --version
```

Configure username.

```
git config --global user.name "Your Name"
```

Configure email.

```
git config --global user.email "yourmail@gmail.com"
```

Verify configuration.

```
git config --list
```

Step 2: Configure Notepad++ as Default Editor

Check whether Notepad++ is installed.

```
notepad++
```

If it is not recognized, add the Notepad++ installation path to the Windows Environment Variables.

Set Notepad++ as the default editor.

```
git config --global core.editor "notepad++"
```

Verify the editor.

```
git config -e
```

Step 3: Create Local Repository

Move to the desired folder.

```
cd /d/D:/GitDemo
```

Initialize Git.

```
git init
```

Check repository status.

```
git status
```

Step 4: Create a File

Create a text file.

```
echo Welcome to Git > welcome.txt
```

Verify the file.

```
dir
```

Display the file contents.

```
cat welcome.txt
```

Step 5: Add File to Git

Track the file.

```
git add welcome.txt
```

Check status.

```
git status
```

Step 6: Commit Changes

Commit the file.

```
git commit -m "Added welcome.txt"
```

Verify the commit.

```
git log
```

Step 7: Connect to GitHub Repository

Create a repository named **GitDemo** in GitHub.

Connect the local repository.

```
git remote add origin https://github.com/YourUsername/GitDemo.git
```

Verify the remote.

```
git remote -v
```

Step 8: Pull Repository

```
git pull origin main --allow-unrelated-histories
```

Step 9: Push Repository

```
git push -u origin main
```

Outputs:

MINGW64:/c/Users/HP/Desktop/GitDemo

```
HP@LAPTOP-JQJ9603M MINGW64 ~
$ git --version
git version 2.50.1.windows.1

HP@LAPTOP-JQJ9603M MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Vanga Gayathri
user.email=gayatri.v192@gmail.com

HP@LAPTOP-JQJ9603M MINGW64 ~
$ cd Desktop

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop
$ mkdir GitDemo

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop
$ cd GitDemo

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo
$ git init
Initialized empty Git repository in C:/Users/HP/Desktop/GitDemo/.git/

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ notepad welcome.txt

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ ls
welcome.txt

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    welcome.txt

nothing added to commit but untracked files present (use "git add" to track)

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git add welcome.txt
```

MINGW64:/c/Users/HP/Desktop/GitDemo

```
$ cd GitDemo
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo
```

```
$ git init
```

```
Initialized empty Git repository in C:/Users/HP/Desktop/GitDemo/.git/
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ notepad welcome.txt
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ ls
```

```
welcome.txt
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ git status
```

```
On branch master
```

```
No commits yet
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
```

```
welcome.txt
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ git add welcome.txt
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ git commit -m "Added welcome.txt"
```

```
[master (root-commit) 5f4fe8f] Added welcome.txt
```

```
1 file changed, 3 insertions(+)
```

```
create mode 100644 welcome.txt
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ git remote add origin https://github.com/GayathriVanga/GitDemo.git
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ git remote -v
```

```
origin https://github.com/GayathriVanga/GitDemo.git (fetch)
```

```
origin https://github.com/GayathriVanga/GitDemo.git (push)
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ git branch
```

```
* master
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ git push -u origin master
```

```
Enumerating objects: 3, done.
```

```
Counting objects: 100% (3/3), done.
```

```
Writing objects: 100% (3/3), 272 bytes | 90.00 KiB/s, done.
```

```
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
```

```
To https://github.com/GayathriVanga/GitDemo.git
```

```
* [new branch]      master -> master
```

```
branch 'master' set up to track 'origin/master'.
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ !
```

GayathriVanga/GitDemo

Ask Gemini

github.com/GayathriVanga/GitDemo

GayathriVanga / GitDemo

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master 1 Branch 0 Tags

Go to file

Add file

Code

GayathriVanga Added welcome.txt 5f4fe8f - 9 minutes ago 1 Commit

welcome.txt Added welcome.txt 9 minutes ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors

No contributors